2. Checking the fit of distributions to data

Quantiles and Quantile function

Recall that the pth **quantile** of the distribution of a random variable X is $Q(p)$ such that

$$P\{X \leq Q(p)\} = p, \quad \text{i.e.} \quad F\{Q(p)\} = p$$

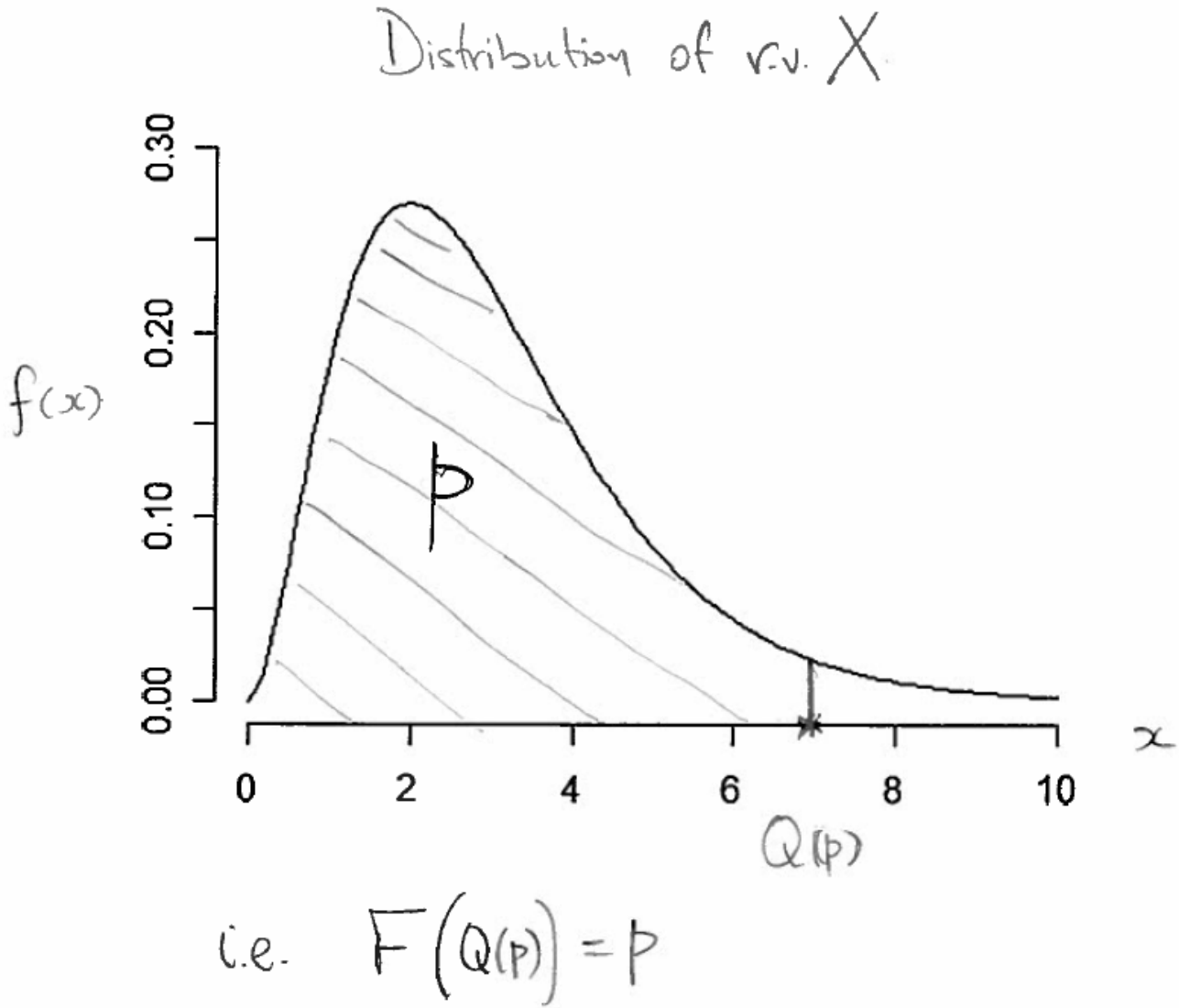
with $F(\cdot)$ being the cdf of X

Then, the (theoretical) **quantile function** $Q(\cdot)$ of a distribution is given by

$$Q(p) = F^{-1}(p)$$

i.e. it is the inverse of the cdf.

So, for example the pth quantile of the distribution of a random variable X is shown schematically below:



Idealised samples

Quantile functions can help produce 'idealised' samples from probability distributions.

First define a sequence of n probability points, equally spaced between 0 and 1, i.e.

$$p_i = \frac{i - 0.5}{n}, \quad i = 1 \ldots, n.$$

For example, for $n = 20$:

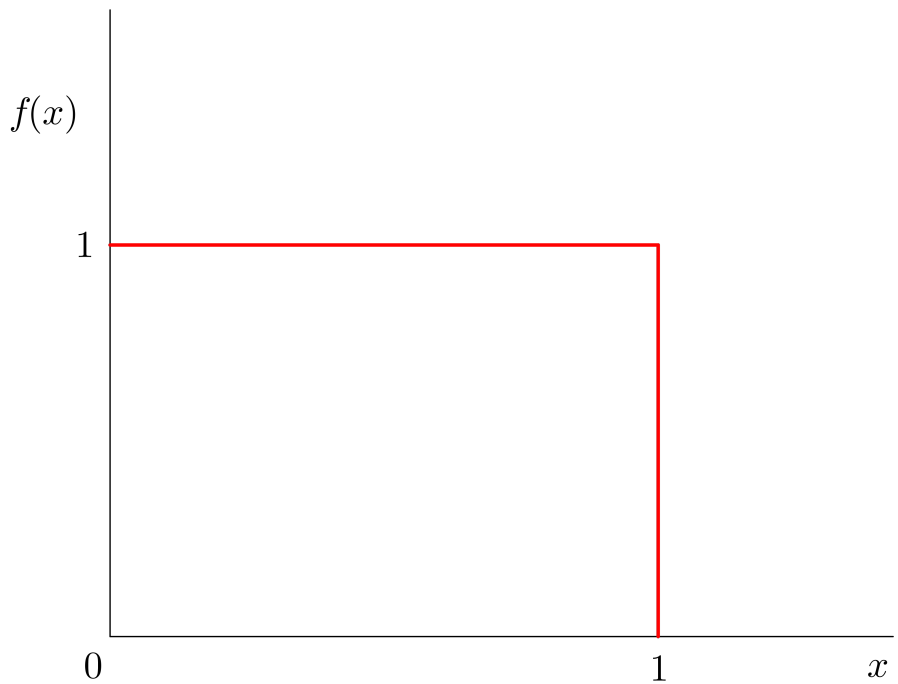
$$p_1 = 0.025, \;\; p_2 = 0.075, \ldots, \;\; p_{20} = 0.975.$$

Then, if a r.v. X has quantile function $Q(\cdot)$, we have (by definition):

$$P\{X \leq Q(p_i)\} = p_i \Rightarrow F\{Q(p_i)\} = p_i, \text{ for } i = 1 \ldots, n$$

and therefore, given $p_1, p_2, \ldots, p_n$, the quantiles $Q(p_1), Q(p_2), \ldots, Q(p_n)$ can be regarded as an idealised sample from the distribution with cdf $F(\cdot)$

The Uniform $U(0, 1)$ Distribution



The Uniform Distribution

Remember

$$F(x) = x \Rightarrow F^{-1}(x) = x \quad \text{i.e.} \quad F(x) = F^{-1}(x)$$

which gives the quantile function as

$$Q(p) = p \quad \text{e.g.} \quad Q(0.2) = 0.2$$

Hence $p_1, \ldots, p_n$ may be regarded as a sample of size n from the $U(0,1)$ distribution.

The Normal $N(0,1)$ Distribution

For $Z \sim N(0,1)$

$$\Phi(z) = p \quad \text{and} \quad Q(p) = \Phi^{-1}(p) = z$$

with

$$\Phi(z) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-x^2/2}\, dx$$

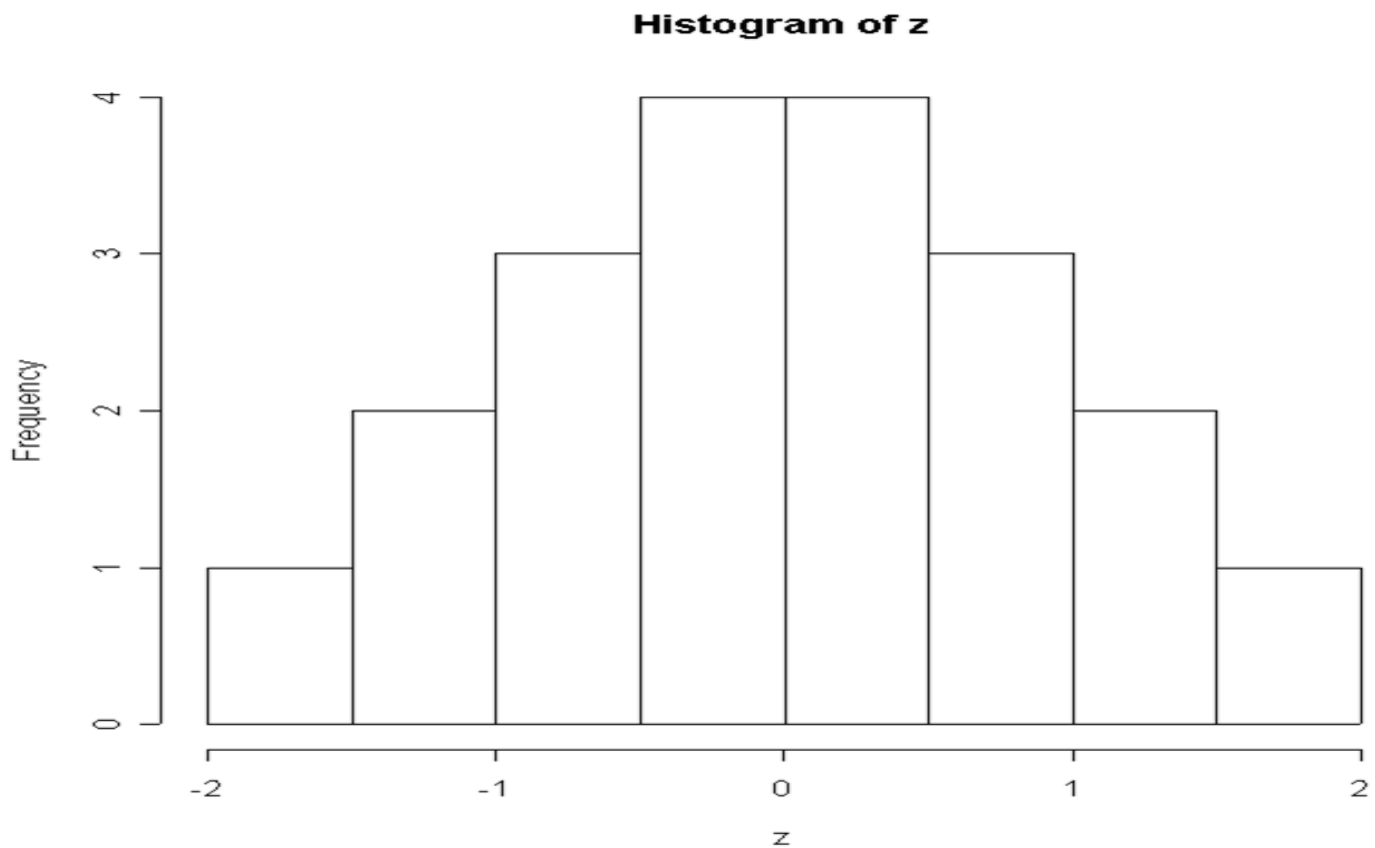
E.g. for $p = 0.025$, we have

$$\Phi(-1.96) = 0.025, \quad \text{so} \quad \Phi^{-1}(0.025) = Q(0.025) = -1.96$$

In R, `qnorm(ppoints(20))` gives the $N(0,1)$ quantiles for 20 equally spaced probability points in $(0,1)$.

```
> z=qnorm(ppoints(20))
> z
 [1] -1.95996398 -1.43953147 -1.15034938 -0.93458929 -0.75541503 -0.59776013
 [7] -0.45376219 -0.31863936 -0.18911843 -0.06270678  0.06270678  0.18911843
[13]  0.31863936  0.45376219  0.59776013  0.75541503  0.93458929  1.15034938
[19]  1.43953147  1.95996398
```

A histogram can be plotted using `hist(z)`



Exponential Distribution

The $Exp(\lambda)$ distribution has quantile function Q given by

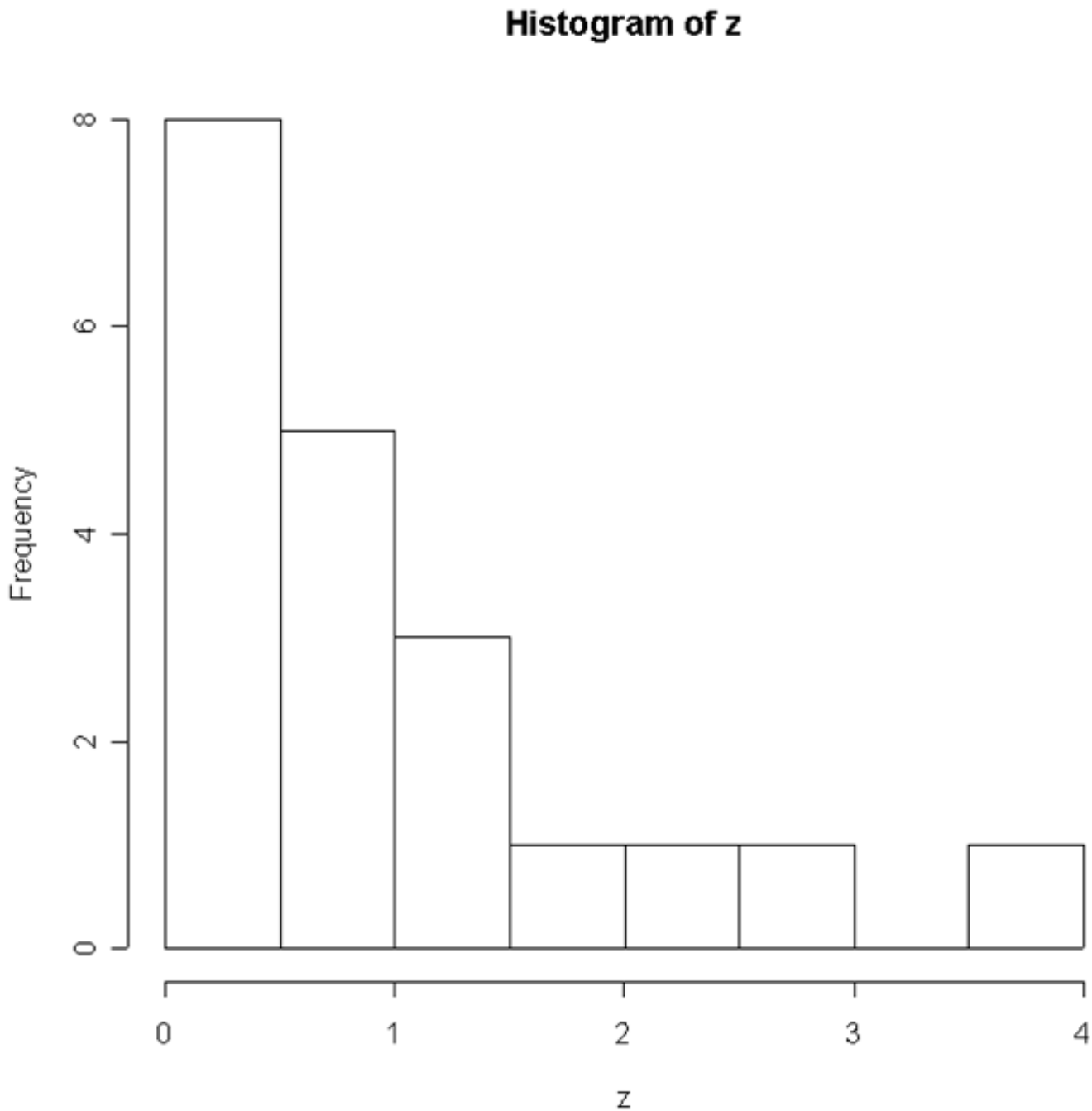
$$Q(p) = -\ln(1-p)/\lambda \qquad \text{(exercise!)}$$

In R it is given by the function `qexp(p, rate = λ)`

(with default $\lambda = 1$).

Thus an $Exp(1)$ sample of size 20, together with its histogram, are obtained as follows:

```
> z=qexp(ppoints(20))
> z
 [1] 0.02531781 0.07796154 0.13353139 0.19237189 0.25489225 0.32158362
 [7] 0.39304259 0.47000363 0.55338524 0.64435702 0.74444047 0.85566611
[13] 0.98082925 1.12393010 1.29098418 1.49165488 1.74296931 2.07944154
[19] 2.59026717 3.68887945
> hist(z)
```


Histogram of z
Frequency
0
2
4
6
8
0
1
2
3
4
z

Empirical Quantile function

Suppose that we have a set of n observations $y_1, \ldots, y_n$ of a variable Y.

Let

$$y_{(1)} \leq \ldots\ldots\ldots \leq y_{(n)}$$

be their **order** statistics (ordered sample).

If we determine $p_1, p_2, \ldots, p_n$ as before, then we define the **empirical quantile function** Q_e of these data by

$$Q_e(p_i) = y_{(i)}$$

(*Note that alternative definitions are also given in the literature*.)

In general, $Q_e(p)$ for other values of p in the interval $(0, 1)$ is given with linear interpolation.

Thus, for any probability p, $Q_e(p)$ is the value of the data that lies above a fraction, p, of the observations.

Remember: the three **quartiles** of the distribution of the data are given by $Q_e(1/4)$, $Q_e(1/2)$ and $Q_e(3/4)$

Example 2.1

Maximum daily ozone concentrations.

The R data frame **ozone** is available on Canvas. It gives maximum daily ozone concentrations, in parts per billion, measured at Stamford, Connecticut and Yonkers, New York, during the period May – September one year.

Data are given only for the 132 days (out of a total of 152) on which readings were successfully obtained at both locations.



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Here ozone is a 'data frame' in R.

We use the command `attach()` to attach names to the variables (columns) of a dataframe.

```
> ozone=dget("ozone.R")
> attach(ozone)
```

```
> ozone
    stamford yonkers
1         66      47
2         52      37
7         49      45
8         64      52
9         68      51
10        26      22
11        86      27
12        52      25
14        75      55
15        87      72
16       188     132
18       103     106
19        82      42
20        71      45
21       103      80
22       240     107
23        31      21
24        40      50
25        47      31
26        51      37
27        31      19
28        47      33
29        14      22
```

R Console

```
128        52        27
129        14         9
130        38        16
131        94        67
132        89        74
133        99        74
134       150        75
135       146        74
136       113        42
138        66        38
139        38        23
140        80        50
141        80        34
142        99        58
143        71        35
144        42        24
145        52        27
146        33        17
147        38        21
148        24        14
149        61        32
150       108        51
151        38        15
152        28        21
> 
```

The sorted Stamford data are given by

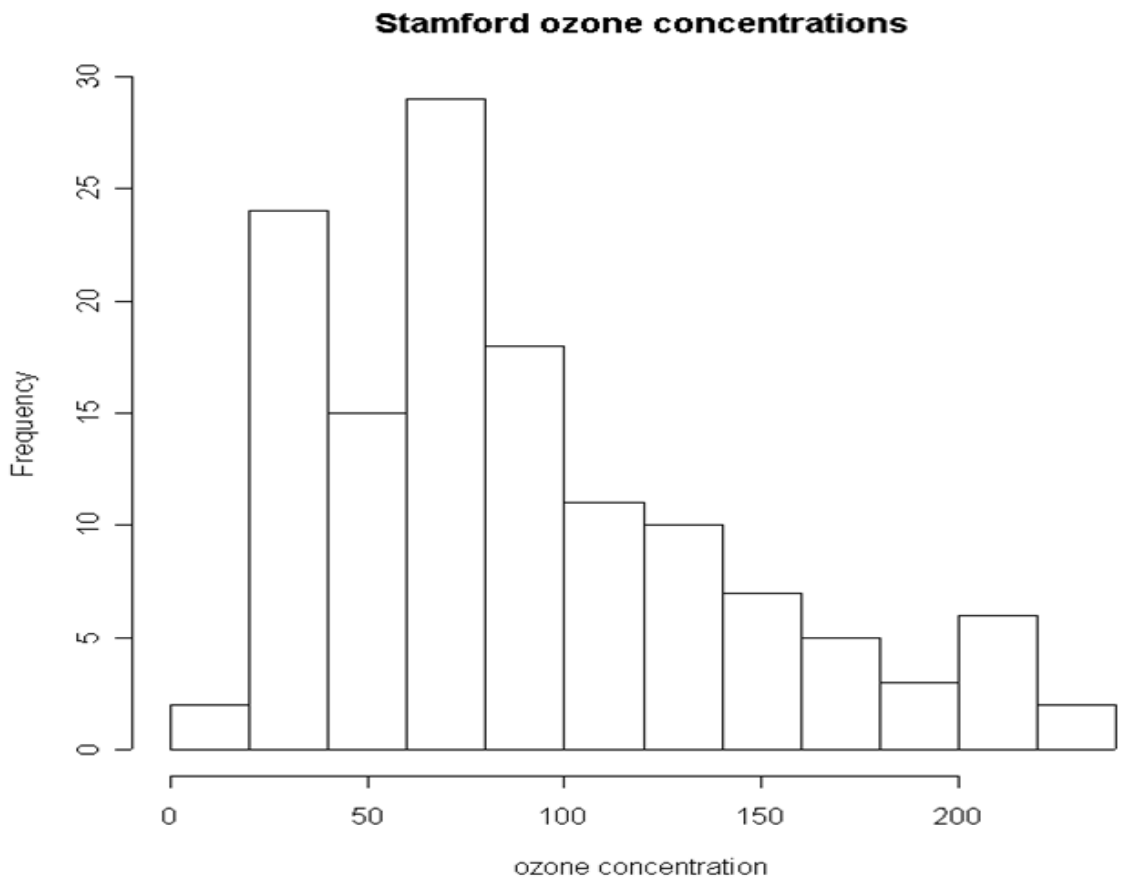
R Console

```
140        80        50
141        80        34
142        99        58
143        71        35
144        42        24
145        52        27
146        33        17
147        38        21
148        24        14
149        61        32
150       108        51
151        38        15
152        28        21
> sort(Stamford)
Error in inherits(x, "factor") : Object "Stamford" not found
> sort(stamford)
  [1]   14   14   23   24   24   24   24   26   27   28   28   28   31   31   32   32   33   37
 [19]   38   38   38   38   38   38   38   40   42   46   47   47   47   47   49   51   52   52
 [37]   52   52   55   59   60   61   61   64   64   64   64   66   66   68   68   68   69   71
 [55]   71   71   71   71   71   72   72   73   75   75   80   80   80   80   80   80   82   82
 [73]   83   85   86   86   87   87   87   89   91   92   94   94   98   99   99  100  101  103
 [91]  103  103  108  111  113  113  114  119  119  122  122  124  124  124  125  125  131  134
[109]  136  141  142  143  146  150  152  155  169  169  170  173  174  188  192  196  201  202
[127]  202  206  212  215  230  240
>
```

The following R code

```
> hist(stamford, xlab="ozone concentration",
main="Stamford ozone concentrations")
```

produces the histogram shown below:



A simple summary is given by

```
summary(stamford)
```

Min	1st Qu.	Median	Mean	3rd Qu.	Max.
14.00	50.50	80.00	89.87	119.80	240.00

Much the same is achieved with the use of the R (empirical) quantile function, which by default evaluates this at

$$p = (0.00, 0.25, 0.50, 0.75, 1.00)$$

```
quantile(stamford)
```

0%	25%	50%	75%	100%
14.00	50.50	80.00	119.80	240.00

Both the histogram and the summaries show the data to be quite positively skewed. This is quite typical of data constrained to be positive.

We can specify any vector of points at which to evaluate the quantile function:

```
>p = seq(0,1,0.1)
>p

[1] 0.0  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```
> p=seq(0,1,0.1)
> quantile(stamford,probs=p)
   0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
 14.0  31.1  42.8  59.3  69.8  80.0  88.2 106.5 125.0 169.9 240.0
```